

SBC 201 Chapter 15 Roof Assembly — Group R-2 High-Rise Dimensional Requirements Matrix (2024)

1. Document metadata and use limitation

- **Project basis:** Riyadh, Saudi Arabia; Group R-2 residential high-rise; an occupied floor is stated to be more than 23 m above the relevant reference level.
- **Deliverable tier:** Project-use matrices in Sections 1–14 (design-check rows, not pasted inventory), plus a coverage summary and unresolved-source register. The full row inventory is not published.
- **Code/source basis:** SBC 201 (2024), Chapter 15, source file Reference\SBC 201 2024\source_reference\Chapter_15 – ROOF ASSEMBLY.txt.
- **Extraction audit:** Skill-finetune run. Project-use rows follow the chapter-extract row contract (noun-phrase checks, bold SI values, building-language triggers, named exceptions, check-specific actions). Internal inventory: **559** independently checkable numeric records (**234** Verified, **325** Verify source). Unresolved OCR is listed in the register and is not a design-release value.
- **Model:** Cursor Grok 4.6.
- **Prepared:** 2026-08-27.
- **Status:** Source-only architectural advisory matrix for design coordination. It is not a stamped compliance statement, roofing specification, SCD NOC, SBPS approval, legal opinion, or permission to construct.
- **Outbound-source rule:** No value in this matrix has been imported from Chapter 16, Section 1611, Section 705.11, Section 1202.2, Chapter 22, Chapter 24, Chapter 26, Table 601, SBC 701, SBC 801, SBC 301, ANSI/SPRI, UL/FM test criteria, commentary examples, or the existing chapter summary. Where Chapter 15 sends the user elsewhere, this matrix records the dependency without supplying the outbound value.

Scope and assumptions

1. Group R-2 and high-rise status are project statements, not independently verified classifications.
2. The exact Riyadh AHJ/permit pathway, project stage, fire-strategy status and SCD NOC status are unconfirmed; therefore this matrix does not conclude compliance.
3. Type of construction, roof covering system, design slope, wind speed/exposure, parapet versus free-edge drainage, ballast/aggregate, occupied-roof/amenity program, penthouse use, cooling towers, PV and vegetative roofs are unconfirmed.
4. Ice-barrier and selected solid-sheathing clauses are printed **Not applicable in the Kingdom**; IBC ice-dam geometry is not filled in.
5. Table 1504.2, Table 1504.9, Table 1505.1, Tables 1507.1.1(1)/(2), Table 1507.3.7 and wood-exposure tables are concatenated OCR. No reconstructed cell is a design-release value.
6. Steep-slope wood, asphalt, slate and clay/concrete tile fastener inventories are omitted from lead tables unless a pavilion or steep roof is added (gap register).

2. Legends

Applicability

Status	Meaning
Direct	Expected to govern the stated R-2 tower basis, subject to confirmed geometry and design data.
Conditional	Governs only when the stated feature, covering, slope, construction type or rooftop structure exists.
Not typical	Unrelated occupancy-only or low-rise covering rule; omitted from this deliverable unless the gap register already opened that use.
External verification	Chapter 15 points to another section/code/standard, or OCR/AHJ must be confirmed before use.

Source confidence

Status	Meaning
Verified	Requirement and any stated numeric value were checked against unambiguous mandatory Chapter 15 source text or an unambiguous table cell.
Verify source	OCR, flattened table, page-split, or footnote attachment is unresolved. Not a design-release value.

3. Project decision and gap register

Decision / gap	Current project basis	Why it controls Chapter 15 application	Required project action
Type of construction	Unconfirmed (high-rise R-2 often Type I, not verified here)	Locks Table 1505.1 class, penthouse height (unlimited on Type I vs 5500 mm otherwise), equipment-screen combustible branches and 1511.2.4 fire-resistance exceptions	Issue a signed construction-type sheet before specifying roof class or penthouse walls
Roof covering and slope	Unconfirmed	Selects 1507.10–1507.14 low-slope membranes vs 1507.4 metal vs steep-slope 1507.2–1507.9; controls 2:12 edge-test and 1/4:12 (2-percent) drainage slopes	Freeze covering system and design slope on the roof plan; do not mix steep-slope fastener tables into a membrane tower by default
Wind speed and exposure	Unconfirmed; Chapter 16 / SBC 301 not imported	Controls Table 1504.2 shingle class, Table 1504.9 aggregate parapets, 1507.1.1 underlayment wind branch and 1507.17.4.1 high-wind caps	Structural engineer to lock V, Vasd, exposure and mean roof height; verify published tables before using OCR cells
Parapet vs free edge	Unconfirmed	Secondary drains/scuppers apply where the perimeter can entrap water; aggregate roofs require Table 1504.9 parapets	Show primary and overflow drainage on every roof catchment; identify parapet/upstand vs dripping eaves
Ballast / aggregate	Unconfirmed	Triggers 1504.5 ANSI/SPRI RP-4, 1507.12.3 stone standards and Table 1504.9 parapet heights	State whether the membrane is adhered, mechanically attached or ballasted
Occupied roof / amenity	Unconfirmed	Weatherproofing, drainage, landscaped-roof fire design and rooftop-structure limits still apply; occupant-load/egress are outside this chapter	Coordinate occupied-roof finishes with 1507.15 / 1505.10 if landscaped; keep Chapter 10/11 geometry out of this matrix
Penthouse program	Unconfirmed	One-third deck-area cap, use limits (MEP/tanks/elevators/shafts only) and Type I FSD rating exceptions	Schedule enclosed rooftop area vs roof deck; do not program habitable amenity as a 1511.2 penthouse
Cooling towers / screens	Unconfirmed	Noncombustible trigger at 23 m² / 4.6 m / roof > 15 m ; screens 5.5 m (Type IA unlimited)	Roof MEP layout to flag tower base area and screen height vs construction type
PV / vegetative	Unconfirmed	1505.8–1505.10 and 1507.15–1507.17 add listing and outbound SBC 801 / 1607.14.2.2 / Table 601	Confirm PV and landscaped-roof scope before specifying fire class and structural fire resistance

Decision / gap	Current project basis	Why it controls Chapter 15 application	Required project action
New vs reroof	Assumed new construction	1512 waives 2-percent slope and 1502.2 overflow only for existing roofs that already drain	Treat 1512 as inactive unless an existing covering is being recovered or replaced
NOC / roof specification	Unconfirmed	Wind tests, listing and SCD acceptance cannot be concluded from Chapter 15 text alone	Engage the qualified local/roofing/fire consultants before design freeze

4. Drainage and overflow

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1502.1	Roof drainage design	Roof drainage design and installation shall comply with this section, Section 1611 and Chapter 11 of SBC 701	All roof assemblies	No numeric drain sizing is stated in this chapter	External verification	Size primary drains from 1611 and SBC 701; do not invent millimetre or rainfall values here	Verified
1502.2	Secondary overflow drainage	Where roof drains are required, secondary (emergency overflow) drains or scuppers are required where perimeter construction can entrap water if primary drains back up; install and size overflow to Section 1611 and SBC 701 Chapter 11	Roof perimeter can pond	Overflow is not required where the roof cannot pond (free-draining edge)	Direct	Provide a second drain path at every parapet or upstand catchment; keep 1611 ponding depth on the structural drawing	Verified
1502.3	Scupper opening	Scuppers sized so ponding does not exceed the 1611.1 design depth; opening dimension not less than 100 mm ; ignore primary-system flow when locating and sizing	Scuppers used as secondary drainage	No other numeric opening stated	Direct	Detail scupper width/height ≥ 100 mm and set inlet elevation from the structural ponding limit, not from the primary drain	Verified
1502.4	External gutters and leaders	Gutters and leaders on the outside of the building shall be noncombustible or minimum Schedule 40 plastic pipe	External gutters/leaders	Group R-3, private garages and Type V construction are excepted	Direct	Specify noncombustible or Schedule 40 plastic leaders on the tower; do not use the R-3/Type V branch	Verified

5. Weather protection and parapet copings

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1503.1	Roof covering required	Roof decks shall be covered with approved roof coverings secured per this chapter, this code and the manufacturer's approved instructions	All roof decks	None stated	Direct	Show a listed covering on every deck, including canopies and low roofs	Verified
1503.2–1503.2.1	Flashing locations and metal	Flash to prevent water entry at copings, permeable materials, parapet intersections and penetrations; flash wall/roof intersections, gutters, slope/direction changes and openings. Metal flashing: corrosion-resistant, thickness not less than 0.48 mm (No. 26 galvanized)	Discontinuities in the roof plane	None stated	Direct	Schedule corrosion-resistant metal flashing ≥ 0.48 mm at every roof/wall, gutter, change-of-slope and penetration	Verified
1503.3–1503.3.2	Parapet coping	Cope or cover parapets so the top surface provides positive drainage. Weatherproof covering width not less than the parapet thickness; rated parapets required by 705.11 shall not have their fire-resistance rating decreased	Parapet walls	1503.3.2 covers parapets that meet a 705.11 exception; still require full-width coping	Direct	Slope the coping to the roof and keep the cap width ≥ wall thickness; coordinate rating with 705.11 without importing that section's heights	Verified
1503.4	Attic and rafter vents	Intake and exhaust vents per Section 1202.2 and the vent manufacturer's instructions	Attic or rafter cavity requiring ventilation	No vent area is stated in this chapter	External verification	Size vents from Chapter 12; do not copy 1202.2 ratios into this matrix	Verified
1503.5	Cricket or saddle	Cricket or saddle on the ridge side of any chimney or penetration greater than 760 mm wide measured perpendicular to the slope; covering of sheet metal or the same material as the roof covering	Penetration or chimney > 760 mm across slope	Unit skylights installed per 2405.5 and flashed per manufacturer may omit the cricket	Conditional	Provide a cricket upslope of wide chimneys, shafts and equipment curbs; record the skylight exception only for listed unit skylights	Verified

6. Wind, edge and impact performance

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1504.1	Wind resistance of roofs	Roof decks and coverings shall be designed for wind loads in accordance with Chapter 16 and 1504.2–1504.5	All roof decks and coverings	No Chapter 16 pressures are stated here	External verification	Put component-and-cladding uplift on the structural/roofing spec; do not import 1609 values	Verified
1504.3.1.3	Clay/concrete tile lift coefficient	Lift coefficient for concrete and clay tile shall be 0.2 , or determined per SBCCI SSTD 11 or ASTM C1570	Clay or concrete roof tile	Used only if tile is selected	Conditional	If tile appears on a pavilion, adopt 0.2 or a tested coefficient; not a default tower membrane check	Verified
1504.4	Mechanically attached or adhered covering	Coverings under 1507 that are mechanically attached or adhered shall resist design wind load pressures for components and cladding per 1609.5.2 ; allowable stress design wind load is permitted	Non-ballasted 1507 covering	Structural metal panels follow 1504.4.2 (ASTM E1592 / FM 4474 / UL 580) with cold-formed steel and aluminium design exceptions to 2210.1 / 2002.1	Direct	Require listed wind-uplift tests on the membrane or metal-panel specification; keep 1609 calculation outbound	Verified
1504.5	Ballasted low-slope single-ply	Ballasted low-slope (roof slope < 2:12) single-ply coverings installed per 1507.12 shall be designed per ANSI/SPRI RP-4	Ballasted single-ply, slope < 2:12	No RP-4 ballast weights are stated here	Conditional	If stone ballast is used, design to RP-4; do not invent bag or stone weights	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1504.6–1504.6.1	Low-slope edge and gutter tests	Metal edge systems (except gutters and counterflashing) on BUR, modified bitumen and single-ply with slope less than 2:12 shall be designed for Chapter 16 wind and tested to ANSI/SPRI ES-1 RE-1, RE-2 and RE-3 (V from Figures 1609.3(1)–(4)). Gutters that secure the membrane edge on the same low-slope systems: Section 1609 plus SPRI GT-1 G-1 and G-2	Low-slope membrane roofs < 2:12	Gutters and counterflashing are excluded from the ES-1 edge-system sentence; GT-1 applies only where the gutter secures the membrane	Direct	Specify ES-1 tested copings/fascia and GT-1 tested gutters on the typical membrane tower edge	Verified
1504.7	Low-slope weathering	Low-slope coverings (slope < 2:12) per 1507 shall demonstrate physical integrity based on 2,000 hours accelerated weathering (ASTM G152, G154 or G155)	Low-slope 1507 covering	Cyclical flexural membranes must not show significant tensile/breaking-strength loss	Direct	Require the 2,000-hour weathering listing on the membrane submittal	Verified
1504.8	Low-slope impact	Low-slope coverings (slope < 2:12) shall resist impact per ASTM D3746, ASTM D4272 or FM 4470 foot-traffic	Low-slope 1507 covering	No impact energy is stated in this chapter	Direct	Require an impact-test listing on the membrane spec; do not import commentary dart masses	Verified
1504.9	Aggregate-surfaced parapet	Parapets shall be provided for aggregate-surfaced roofs and shall comply with Table 1504.9	Aggregate-surfaced roofs	Table grid is concatenated OCR — no millimetre row adopted	Conditional	If gravel/slag surfacing is used, provide a parapet and verify the published table before setting height	Verified
Table 1504.9 notes c–d	Gravel stop and Exposure D	Where the table height is 50 mm , a gravel stop ≥ 50 mm from the roof surface and ≥ aggregate height is permitted. Exposure D: add 200 mm to Exposure C height, and not less than 300 mm	Aggregate roofs using Table 1504.9	Interpolation of height and wind is permitted (note a); V and exposure from 1609 (note b)	Conditional	Use these footnote rules only after the published table cell is verified; do not reconstruct the OCR grid	Verified
Table 1504.2	Steep-slope shingle wind class	Classification vs maximum basic / allowable-stress wind speed cannot be unambiguously segmented from the concatenated table	Asphalt, metal or PV shingles tested to ASTM D3161 / D7158 / UL 7103	Footnote a: ASTM D7158 calculations assume Exposure B or C and building height 18 m or less ; additional calculations outside those assumptions	External verification	Verify the published table before assigning a D/G/H or A/D/F class; high-rise height typically exceeds the 18 m footnote assumption	Verify source
Table 1504.9	Aggregate parapet height grid	Minimum parapet height (mm) vs aggregate size, mean roof height and wind cannot be read from the flattened HTML	Aggregate-surfaced roofs	Notes c–d above remain usable once the cell is known	External verification	Hold all millimetre grid values until a published Table 1504.9 is checked	Verify source

7. Fire classification

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1505.1	Roof-assembly class test	Class A, B and C assemblies required to be listed shall be tested to ASTM E108 or UL 790; fire-retardant-treated wood roof coverings also ASTM D2898. Minimum covering class per Table 1505.1 by construction type	Roof assemblies	Skylights and sloped glazing complying with Chapter 24 or Section 2610 are excepted from this fire classification	Direct	Specify a listed Class A/B/C assembly matching construction type; keep skylight fire rules on Ch. 24 / 2610	Verified
1505.2	Class A use and deemed-to-comply	Class A assemblies are effective against severe fire-test exposure, shall be listed/identified Class A, and are permitted on all types of construction . Exceptions (no listing required): brick, masonry or exposed concrete; ferrous/copper/metal sheets and shingles, clay/concrete tile or slate on noncombustible decks (or metal sheets without a deck on noncombustible framing); copper sheets minimum 4.9 kg/m² over combustible decks; slate over ASTM D226 Type II over combustible decks	Class A covering	Exceptions 1–4 are deemed Class A without ASTM E108 listing	Direct	Prefer a listed Class A membrane on the tower; if using copper over combustible deck, schedule ≥ 4.9 kg/m²	Verified
1505.3–1505.5	Class B, C and nonclassified	Class B: moderate exposure, listed Class B. Class C: light exposure, listed Class C. Nonclassified: approved material not listed as A, B or C	Coverings other than Class A	Table 1505.1 (not commentary) sets the minimum class by construction type	Direct	Do not assume Class C is available on a high-rise Type I plate; verify Table 1505.1	Verified
Table 1505.1	Minimum class by construction type	Appended table is concatenated (IA...VB with BBBC°BC°BBC° tokens). No construction-type class letter is adopted	All buildings	Notes below are readable; Urban Wildland / Appendix D may impose a stricter class	External verification	Verify the published Table 1505.1 against the locked construction type before specifying Class B vs C	Verify source
Table 1505.1 notes	Nonclassified and cedar-shake limits	Nonclassified coverings permitted on Group R-3 and U with fire-separation distance ≥ 1.8 m from the leading edge of the roof. Cedar/redwood No. 1 shakes/shingles per 1505.7: not more than two stories , projected roof ≤ 560 m² , FSD ≥ 3.0 m to lot lines except street/public way	Those limited buildings only	Not a Group R-2 high-rise path	Not typical	Do not apply R-3/U nonclassified or two-storey cedar notes to this tower	Verified
1505.7	Special-purpose wood underlayment	Underlayment of 16 mm Type X water-resistant gypsum backing board or gypsum sheathing under minimum nominal 12.5 mm wood structural panel solid sheathing or 25 mm nominal spaced sheathing	Special-purpose wood shingle/shake roofs per 1507.8 / 1507.9	Buildings permitted to use this construction are those in Table 1505.1 note (two-storey / 560 m²)	Not typical	Omit from the tower covering spec unless a qualifying special-purpose wood roof is added	Verified

8. Low-slope coverings typical of towers

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1507.10.1	Built-up roof slope	Design slope not less than one-fourth unit vertical in 12 units horizontal (2-percent slope) , except coal-tar built-up not less than one-eighth unit vertical in 12 units horizontal (1-percent slope)	Built-up roofs	Coal-tar branch is the only lower-slope exception in this clause	Direct	Set BUR drainage slope to 2-percent unless a coal-tar specification is locked at 1-percent	Verified
1507.10.2	Built-up materials	BUR materials shall comply with Table 1507.10.2 or UL 55A	Built-up roofs	Table 1507.10.2 is a standards list (no dimensional cells)	Direct	Call the applicable ASTM/UL 55A standard on the BUR spec	Verified
1507.11.1–1507.11.2	Modified-bitumen slope and materials	Design slope not less than one-fourth unit vertical in 12 units horizontal (2-percent slope) . Materials: ASTM D6162, D6163, D6164, D6222, D6223, D6298 or D6509. Base sheet may be 1507.11.2, ASTM D1970 or ASTM D4601	Modified-bitumen roofs	None stated	Direct	Set mod-bit slope to 2-percent and list a named ASTM product	Verified
1507.12.1–1507.12.2	Single-ply slope and materials	Design slope not less than 1/4 unit vertical in 12 units horizontal (2-percent slope) . Materials per Table 1507.12.2 (CSPE/PIB, EPDM, KEE, PVC, TPO)	Single-ply roofs	None stated	Direct	Set single-ply slope to 2-percent and name the ASTM membrane	Verified
1507.12.3	Ballasted stone	Ballasted low-slope roofs (slope < 2:12) installed per this section and 1504.5; stone ballast ASTM D448 or ASTM D7655	Ballasted single-ply < 2:12	RP-4 design remains outbound	Conditional	If ballast is used, specify ASTM D448/D7655 stone and the 1504.5 RP-4 design	Verified
1507.13.1–1507.13.4	SPF slope, coating window and foam	SPF design slope not less than one-fourth unit vertical in 12 units horizontal (2-percent slope) . Foam ASTM C1029 Type III or IV or ASTM D7425. Liquid-applied protective coating per Table 1507.13.3 applied not less than 2 hours nor more than 72 hours after foam. Foam plastics also Chapter 26	Sprayed polyurethane foam roofs	Chapter 26 values are not imported	Conditional	If SPF is used, hold 2-percent slope and the 2–72 hour coating window on the application spec	Verified
1507.14.1–1507.14.2	Liquid-applied slope	Liquid-applied roofing design slope not less than one-fourth unit vertical in 12 units horizontal (2-percent slope) ; ASTM C836, C957 or D3468	Liquid-applied roofs	None stated	Conditional	If liquid-applied, set slope to 2-percent and name the ASTM coating	Verified
1507.4.2	Metal panel minimum slope	Lapped non-soldered seams without lap sealant: three units vertical in 12 (25-percent slope) . With approved lap sealant: one-half unit vertical in 12 (4-percent slope) . Standing-seam: one-quarter unit vertical in 12 (2-percent slope)	Metal roof panels	Deck may be spaced supports where the covering is so designed (1507.4.1)	Conditional	If metal panels are used, pick the matching slope branch; standing-seam at 2-percent is the low-slope tower option	Verified
1507.4.3	Metal covering standards	Metal-sheet systems with supporting members: Chapter 22 . Coverings over structural decking: Table 1507.4.3(1); corrosion resistance Table 1507.4.3(2)	Metal-sheet roof coverings	Table 1507.4.3(1) copper masses 0.0416 kg/m² / 0.0312 kg/m² are treated as OCR-suspect	Conditional	Use readable aluminium 0.61 mm / 0.48 mm , zinc 0.7 mm and lead masses from the table; verify copper masses against the published table	Verify source
1506.3	Product identification	Roof-covering materials delivered in packages bearing manufacturer marks and approved-agency labels required by 1505; bulk shipments by certificate or bill of lading	All 1506/1507 materials	None stated	Direct	Require labeled bundles/rolls on site and in the submittal	Verified

9. Insulation, coatings and radiant barriers

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1508.1	Above-deck insulation fire test	Above-deck thermal insulation permitted where covered with an approved roof covering and the assembly passes NFPA 276 or UL 1256	Above-deck insulation	(1) Foam plastic per Chapter 26 . (2) Concrete or composite metal-and-concrete deck with approved covering — NFPA 276/UL 1256 not required for that non-foam case	Direct	Require NFPA 276 or UL 1256 on steel-deck insulation assemblies; send foam to Chapter 26; document the concrete-deck exception if used	Verified
1508.2	Insulation material standards	Above-deck thermal insulation board shall comply with Table 1508.2	Above-deck insulation board	Table is a standards list	Direct	Name the ASTM C552 / C578 / C1289 (etc.) product on the insulation spec	Verified
1509.1–1509.2	Roof coatings	Installing a roof coating on a covering shall comply with 1505 and this section; materials per Table 1509.2	Roof coatings	Coatings still need a 1505 fire class where required	Conditional	If recoating, list ASTM D6083 / D1227 / D2823 / D4479 / D2824 / D6694 / D6947 as applicable and keep 1505 class	Verified
1510.2–1510.4	Above-deck radiant barrier	Permitted where covered with an approved covering and the radiant-barrier-plus-covering system complies with FM 4550 or UL 1256 . Low-emittance surface shall face the continuous airspace between barrier and covering. Material: ASTM C1313/1313M	Radiant barrier above the deck	None stated	Conditional	If used, orient the low-e face to the airspace and require FM 4550 or UL 1256 on the combined system	Verified

10. Rooftop structures

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1511.1.1	Enclosed rooftop area	Aggregate area of penthouses and other enclosed rooftop structures shall not exceed one-third the area of the supporting roof deck. Such structures need not be included in building area or storey count per 503.1 , nor in the 901.7 fire area	Penthouses and enclosed rooftop structures	Exceeding one-third means the enclosure is treated as an additional storey (per the code's penthouse definition path in 1511.2)	Direct	Schedule enclosed roof area vs deck area and keep the sum ≤ one-third	Verified
1511.2–1511.2.2	Penthouse height and use	Penthouses complying with 1511.2.1–1511.2.4 are a portion of the storey below; others are an additional storey. Height above roof deck, measured to average penthouse roof: not more than 5500 mm except on Type I (unlimited). Use limited to shelter of mechanical/electrical equipment, tanks, elevators and related machinery, or vertical shaft openings, including ancillary access to elevators and stairways	Penthouses	Exception: tanks or elevators that travel to the roof — maximum 8500 mm on non-Type I	Direct	On a Type I tower, height is unlimited but use remains MEP/shafts only; do not occupy a 1511.2 penthouse as amenity	Verified
1511.2.3	Penthouse weather protection	Louvers, louver blades or flashing shall protect mechanical/electrical equipment and the building interior from the elements	Penthouses with openings	None stated	Direct	Detail weatherproof louver/flashing at penthouse intakes and doors	Verified
1511.2.4 Ex. 1	Type I penthouse rating vs FSD	Type I penthouses: exterior walls and roofs with fire separation distance > 1.5 m and < 6 m may have not less than a 1-hour rating; FSD 6 m or greater — no fire-resistance rating required	Type I construction penthouses	Base rule: penthouses constructed as required for the building's construction type	Conditional	Apply the 1.5 m / 6 m / 1-hour branch only after construction type and FSD are locked	Verified
1511.2.4 Ex. 2	Low Type I or Type II penthouse FRT	On Type I two stories or less above grade plane, or Type II: FSD > 1.5 m and < 6 m — not less than 1-hour (or lesser per Table 705.5) and FRT wood permitted; FSD ≥ 6 m — FRT wood, no rating; interior framing/walls may be FRT wood	Type I ≤ 2 stories or Type II	Not the high-rise Type I path	Not typical	Do not use the two-storey/Type II FRT branch on a high-rise Type I tower	Verified
1511.2.4 Ex. 3	Type III/IV/V penthouse walls	Exception 3 text is OCR-repeated and page-split; remaining limits are not adopted	Type III, IV or V penthouses	Not readable	External verification	Verify published 1511.2.4 Exception 3 before any Type III–V penthouse wall rating	Verify source
1511.3–1511.3.2	Roof tanks	Clauses 1511.3, 1511.3.1 and 1511.3.2 are missing after the page 1487–1488 split (next numbered code is 1511.3.3)	Roof tanks	Commentary about tanks over stairs/elevators is not adopted	External verification	Verify published 1511.3–1511.3.2 before locating tanks; do not use commentary as the requirement	Verify source
1511.3.3	Unenclosed tank cover	Unenclosed roof tanks shall have covers sloping toward the perimeter of the tanks	Unenclosed roof tanks	None stated	Conditional	Slope the tank cover to shed rain at the perimeter	Verified
1511.4	Cooling-tower construction	Cooling towers on the roof with base area > 23 m² or height > 4.6 m above the roof, where the roof is > 15 m above grade plane, shall be noncombustible. Base area shall not exceed one-third of the supporting roof deck	Roof cooling towers meeting the size/height/roof-height triggers	Exception: drip boards and enclosing construction may be wood not less than 25 mm nominal if covered on the exterior with noncombustible material	Conditional	If a tower exceeds 23 m² or 4.6 m on a roof > 15 m , specify noncombustible construction and keep base area ≤ one-third	Verified
1511.5	Towers, spires, domes, cupolas	Construction type fire-resistance not less than the building. If > 26 m above grade plane to the highest point and either > 18.6 m² horizontal area or used for other than a minaret or architectural embellishment: construct of and support on Type I or II	Towers/spires/domes/cupolas	Minaret / architectural-embellishment use avoids the Type I/II trigger when the area/height pair is not met	Conditional	Classify rooftop minarets vs occupied towers; if both 26 m and 18.6 m² /non-embellishment apply, require Type I or II	Verified
1511.5.1	Noncombustible tower separation	Greater than 18 m above roof contact, or > 18.6 m² at any horizontal section, or used other than minaret/embellishment, or on a building > 1.5 m in building height: noncombustible construction and support, separated by not less than 1.5 hours with openings per 711 . On a building > 15 m in building height: supported by noncombustible construction	Qualifying towers/spires/domes/cupolas	The 1.5 m building-height token sits beside a 15 m token in the same paragraph and is treated as unresolved OCR	Conditional	Use 18 m , 18.6 m² , 1.5-hour and 15 m as printed; verify the 1.5 m building-height trigger against the published clause	Verify source
1511.6–1511.6.1	Mechanical equipment screens	Screens constructed of exterior-wall materials for the building's construction type. FSD > 1.5 m : fire-resistance rating not required. Height not more than 5.5 m above the roof deck, except unlimited on Type IA	Mechanical equipment screens	Fences (1511.7.4) use the same rules	Direct	Limit screen height to 5.5 m unless Type IA; drop the rating when FSD > 1.5 m	Verified
1511.6.2	Combustible screens on I–IV	On Type I, II, III or IV roofs, combustible screens permitted if any one of: (1) FSD ≥ 6 m and height ≤ 1200 mm ; (2) FSD ≥ 6 m and FRT wood for exterior installation; (3) panels flame spread index 25 or less (ASTM E84 or UL 723, each face) and NFPA 285 as tested	Combustible screen on Type I–IV	Item 3 requires installation as tested	Conditional	If a combustible screen is proposed on the tower, pick one numbered limitation and show FSD/height/FRT/NFPA 285 on the detail	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1511.6.3	Type V screen height	Type V screens may exceed maximum building height where FSD > 1500 mm and one of: FSD ≥ 6 m with 1200 mm extra height; noncombustible; FRT wood; or FSD ≥ 6 m with flame spread 25 or less	Type V buildings	Not the typical high-rise construction type	Not typical	Do not apply Type V height increases to a Type I tower	Verified
1511.7.1	Aerial supports	Aerial supports shall be noncombustible, except combustible permitted where height not greater than 3650 mm from roof deck to the highest point	Aerial supports	Height exception as stated	Conditional	Keep aerials noncombustible, or limit combustible supports to ≤ 3650 mm	Verified
1511.7.2	Bulkheads	Bulkheads sheltering MEP or shaft openings comply as penthouses (1511.2); any other purpose is an additional storey	Bulkheads	None stated	Conditional	Treat stair/MEP bulkheads as 1511.2 penthouses	Verified
1511.7.5	Flagpoles	Flagpoles and similar structures need not be noncombustible and are not limited in height or number	Flagpoles	None stated	Conditional	Flagpoles are exempt from 1511 height/combustibility caps	Verified
1511.8	Roof supporting rooftop structures	Structural frame and roof construction supporting rooftop-structure loads shall comply with Table 601 . The fire-resistance reduction in Table 601 Note a shall not apply to roofs containing rooftop structures	Any rooftop structure on the roof	Table 601 hours are not imported	External verification	Tell the structural/fire pack that Note a reduction is off; verify Table 601 hours outside this chapter	Verified

11. PV, BIPV and landscaped roofs

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1505.8	BIPV fire class	BIPV products installed as the roof covering shall be tested, listed and labeled for fire classification per 1505.1	BIPV as covering	ASTM E108 / UL 790 path via 1505.1	Conditional	If BIPV is the covering, require a 1505.1 fire-class listing	Verified
1505.9	Rooftop PV fire class	Rooftop-mounted PV panel systems tested, listed and identified with a fire classification per UL 2703 ; install per listing and manufacturer. Fire classification shall comply with Table 1505.1	Rooftop-mounted PV	Table 1505.1 cells remain Verify source	Conditional	Require UL 2703 class matching the verified Table 1505.1 construction-type minimum	Verified
1505.10	Landscaped-roof fire design	Landscaped roofs shall comply with 1505.1 and 1507.15 and be installed per ANSI/SPRI VF-1	Landscaped roofs	VF-1 criteria are not imported	Conditional	If a roof garden is provided, specify ANSI/SPRI VF-1 and 1505.1 class	Verified
1507.15–1507.15.1	Vegetative / landscaped roof	Vegetative and landscaped roofs shall comply with this chapter, Section 1607.14.2.2 and SBC 801 . Supporting frame and roof construction shall comply with Table 601	Vegetative or landscaped roofs	No live-load or 801 values imported	Conditional	Coordinate structural live load (1607.14.2.2), SBC 801 and Table 601 without copying those numbers here	Verified
1507.16.2 / 1507.16.5	Photovoltaic shingles	Install on slopes not less than two units vertical in 12 (2:12) . Fasteners: galvanized, stainless, aluminium or copper roofing nails, minimum 12-gage (2.67 mm) shank, 9 mm minimum head, penetrating not less than 19 mm into sheathing (through-sheathing if thinner)	Photovoltaic shingles	Ice barrier not applicable in the Kingdom	Conditional	If PV shingles are used, hold 2:12 slope and the 2.67 mm / 9 mm / 19 mm nail rule	Verified
1507.16.8	PV shingle wind class	Photovoltaic shingles shall comply with Table 1504.2 for the appropriate maximum nominal design wind speed	Photovoltaic shingles	Table 1504.2 is concatenated OCR	External verification	Verify published Table 1504.2 before assigning the shingle class	Verify source
1507.17.2–1507.17.4.1	BIPV panel slope and high-wind underlayment	BIPV panels only on slopes two units vertical in 12 (2:12) or greater . Underlayment shingle-fashion from the eave, lapped 50 mm . Where Vad > 49 m/s : overlap fasteners not more than 900 mm oc. Where Vad ≥ 54 m/s : ASTM D226 Type III, D4869 Type IV or D6757; grid 300 mm with 150 mm at side laps; all laps not less than 100 mm ; cap nails/staples with head ≥ 25 mm , metal cap ≥ 0.25 mm , plastic cap edge ≥ 0.89 mm , ring-shank ≥ 2.11 mm , smooth-shank ≥ 2.31 mm , staple ≥ 21 gage (0.81 mm) , penetration through sheathing or ≥ 19 mm	BIPV roof panels	Adhered underlayment ASTM D1970 is an alternative. Ice barrier not applicable in the Kingdom	Conditional	If BIPV panels are used, lock 2:12 and apply the Vad 49 / 54 m/s underlayment branch from Chapter 16 Vasd	Verified

12. Kingdom deletions and reroofing

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1507.1.2 and siblings	Ice barrier	Printed Not applicable in the Kingdom (1507.1.2, 1507.2.7, 1507.5.4, 1507.6.4, 1507.7.4, 1507.8.4, 1507.9.4, 1507.16.4, 1507.17.4.2)	Steep-slope underlayment ice-dam clauses	Do not import IBC ice-barrier millimetres	Direct	Omit ice-barrier strips from KSA specifications; do not fill IBC eave-ice geometry	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1507.8.1.1 / 1507.9.1.1	Solid sheathing for ice	Printed Not applicable in the Kingdom	Wood shingle/shake ice-area solid sheathing	None	Direct	Do not require the IBC ice-area solid-sheathing extra	Verified
Table 1507.3.7 note b	Tile attachment climate note	Printed Not applicable to KSA	Clay/concrete tile Table 1507.3.7	Remainder of the table is flattened OCR	Conditional	If tile is used, verify the published table; do not apply note b	Verified
1512.1 Ex. 1	Existing low-slope recover slope	Roof replacement or recover of existing low-slope coverings need not meet the one-quarter in 12 (2-percent slope) in 1507 where the roof provides positive roof drainage	Existing low-slope reroof	New construction still uses 1507 slopes	Conditional	Use only on existing covers that already drain; do not waive 2-percent on new tower roofs	Verified
1512.1 Ex. 2	Existing overflow waiver	Recovering or replacing an existing covering need not meet 1502.2 secondary drains/scuppers where the roof provides positive drainage; existing required overflow shall not be removed unless replaced to 1502.2	Existing reroof with positive drainage	New construction still needs 1502.2 where the perimeter can entrap water	Conditional	Do not delete existing overflow on a reroof unless a 1502.2 replacement is installed	Verified
1512.2	Roof replacement	Roof replacement shall include removal of all existing layers of roof assembly materials down to the roof deck	Roof replacement	Exception printed Not applicable in the Kingdom	Conditional	Tear off to deck on replacement; do not use a Kingdom-deleted remaining-layer exception	Verified
1512.2.1.1	Recover prohibited	Roof recover not permitted where the existing roof is water-soaked/deteriorated, is slate/clay/cement/asbestos-cement tile, or has two or more applications of any covering	Proposed recover	1512.2.1 lists conditions where recover is otherwise permitted	Conditional	If recovering, confirm fewer than two existing coverings and a sound, dry substrate	Verified

13. Steep-slope coverings (only if used)

Steep-slope asphalt, tile, metal shingle, mineral-surfaced roll, slate, wood shingle and wood shake systems are not typical of a Group R-2 high-rise membrane roof. The following rows apply only if the gap register opens a pavilion, canopy or steep roof. Fastener-count tables and weather-exposure grids that are OCR-flattened are not dumped here.

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1507.2.2	Asphalt shingle slope	Asphalt shingles only on slopes 2:12 (17-percent) or greater; 2:12 to 4:12 (33-percent) requires double underlayment per 1507.1.1	Asphalt shingles	Underlayment tables 1507.1.1(1)–(2) are flattened OCR	Conditional	If asphalt is used, hold 2:12 minimum and double underlayment below 4:12 ; verify published underlayment tables	Verified
1507.2.5–1507.2.6	Asphalt fasteners	Nails: 12-gage (2.7 mm) shank, 9.5 mm minimum head, 19 mm into sheathing (through if thinner). Not fewer than four fasteners per strip shingle or two per individual shingle; slopes exceeding 21:12 per manufacturer	Asphalt shingles	Manufacturer may require more fasteners, not fewer	Conditional	Put four /strip and 19 mm penetration on the asphalt detail	Verified
1507.2.8.3	Asphalt drip edge	Drip edge at eaves and rakes; adjacent segments lapped not less than 50 mm ; vertical leg not less than 38 mm and not less than 6.5 mm below sheathing; back on the roof not less than 50 mm ; fastened not greater than 300 mm oc. Underlayment over drip edge at eaves; drip edge over underlayment at rakes	Asphalt shingle eaves and rakes	None stated	Conditional	Detail the 50 / 38 / 6.5 / 50 / 300 mm drip-edge geometry	Verified
1507.3.2	Clay/concrete tile slope	Tile on slopes 2-1/2:12 (21-percent) or greater; 2-1/2:12 to 4:12 (33-percent) requires double underlayment	Clay or concrete tile	Solid sheathing required except spaced lumber in SDC A, B and C (1507.3.1)	Conditional	If tile is used, hold 2½:12 and double underlayment below 4:12	Verified
1507.3.6	Tile fasteners and perimeter	Corrosion-resistant fasteners, not less than 11-gage, 8.0 mm head, penetrating the deck not less than 19 mm or through, whichever is less. Attaching wire not smaller than 2.1 mm . Perimeter fastening: three tile courses but not less than 900 mm from hips, ridges, eaves and gable rakes	Clay or concrete tile	Table 1507.3.7 fastening schedule is flattened OCR	Conditional	Use the 8.0 mm / 19 mm / 2.1 mm / 900 mm prose rules; verify published Table 1507.3.7 before wind-speed fastening	Verified
1507.5.2 / 1507.5.7	Metal shingle slope and valley	Metal shingles not below 3:12 (25-percent slope) . Valley flashing at least 200 mm each way from centreline, splash rib not less than 19 mm , end lap not less than 100 mm ; cement underlayment or self-adhered sheet for slopes under 7:12 (58-percent)	Metal roof shingles	None stated	Conditional	If metal shingles are used, hold 3:12 and the 200 / 19 / 100 mm valley	Verified
1507.6.2	Mineral-surfaced roll slope	Not applied on slopes below one-unit vertical in 12 (8-percent slope)	Mineral-surfaced roll roofing	None stated	Conditional	If roll roofing is used, hold 8-percent minimum slope	Verified
1507.7.2 / 1507.7.6	Slate slope and headlap	Slate only on slopes 4:12 or greater; two fasteners per slate; headlap per Table 1507.7.6 (100 mm / 75 mm / 50 mm by slope band)	Slate shingles	Table 1507.7.6 slope/headlap tokens are readable	Conditional	If slate is used, hold 4:12 , two fasteners, and the table headlap	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1507.8.2 / 1507.8.7	Wood shingle slope and spacing	Wood shingles on slopes not less than 3:12 (25-percent) . Side lap not less than 40 mm ; spacing 6 to 9 mm ; weather exposure not more than Table 1507.8.7	Wood shingles	Table 1507.8.7 is rowspan-concatenated OCR; table spacing 6.4 to 9.5 mm conflicts with the clause	Conditional	Use clause 6 to 9 mm and 40 mm side lap; verify published Table 1507.8.7 before exposure	Verify source
1507.9.2 / 1507.9.8	Wood shake slope and spacing	Wood shakes on slopes not less than 4:12 (33-percent) . Side lap not less than 40 mm ; spacing 9 to 16 mm (naturally durable) or 6 to 9 mm (preservative taper sawn); exposure per Table 1507.9.8	Wood shakes	Table 1507.9.8 is concatenated OCR	Conditional	Use the clause spacing limits; verify published Table 1507.9.8 before exposure	Verify source
Tables 1507.1.1(1)–(2)	Underlayment type and application	Type/application vs covering and V 62.5 m/s cannot be unambiguously read (headers show both V > and V ≥ 62.5 m/s ; cells concatenated)	Steep-slope underlayment	1507.1.1 Exception 2 grid 300 mm / 150 mm / laps 100 mm / offset 1800 mm remains readable in the exception prose	External verification	Prefer the 1507.1.1 exception prose for cap-nail underlayment; verify published Tables 1507.1.1(1)–(2)	Verify source

14. Project-use controls

1. Use **Verified** rows for initial coordination after the row trigger and branch are confirmed.
2. Treat every **Verify source** row as a design hold point; no affected OCR table cell is to be placed in issued-for-approval drawings without a published-source check.
3. Do not reconstruct Table 1505.1 class letters or Table 1504.9 millimetre grids from commentary, CS.md or memory.
4. Do not import Section 1611 rainfall, SBC 701 drain sizing, 705.11 parapet fire heights, 1202.2 vent ratios, Chapter 16 pressures, Table 601 hours, or SBC 801 landscaped-roof values.
5. Do not apply R-3/U nonclassified roofs, two-storey cedar-shake notes, Type V screen-height increases, or ice-barrier geometry to this R-2 high-rise.
6. Record covering system, construction type, wind and rooftop-structure decisions in the project Golden Thread; this matrix is not evidence of SCD NOC or stamped compliance.

15. Coverage summary

Internal inventory of the attached Chapter 15 extract (numbered code, exceptions, tables, footnotes; commentary excluded). Row-level records are not published.

- **Inventory scope:** numbered code, exceptions, tables, footnotes (commentary excluded)
- **Total independently checkable numeric records:** 559
- **Verified:** 234
- **Verify source:** 325
- **Numeric records in Sections 1501, 1506, 1508, 1509 and 1510:** 0 (1508–1510 name tests/standards without independent SI limits)

Counts by top-level section

Top-level section	Records
1501	0
1502	2
1503	2
1504	211
1505	17
1506	0
1507	288
1508	0
1509	0
1510	0
1511	37
1512	2

Appended-table coverage

Appended table	Records	Verify source records
Table 1504.2	33	32
Table 1504.9	171	168
Table 1505.1	13	9
Table 1507.1.1(1)	1	1
Table 1507.1.1(2)	36	36
Table 1507.1.1(3)	2	0
Table 1507.2.8.2	16	0
Table 1507.3.7	48	35

Appended table	Records	Verify source records
Table 1507.4.3(1)	17	5
Table 1507.4.3(2)	1	1
Table 1507.7.6	3	0
Table 1507.8 (spacing conflict)	2	2
Table 1507.8.7	18	18
Table 1507.9.8	15	12
Tables 1507.10.2 / 1507.12.2 / 1507.13.3 / 1508.2 / 1509.2	0	0

Coverage cross-check against SBC 201 Chapter 15 Roof Assembly (2024)_CS.md was topics-only: roof class, parapets, drainage/scuppers, outbound 1611 / SBC 701 / 705.11. No CS.md value was copied into a matrix cell.

16. Unresolved-source register

Hold points for the 325 **Verify source** inventory records. Counts are record counts, not distinct numeric values. No value in this register is a design-release figure.

Affected table / clause	Why unverified	Control note
Table 1504.2	32 concatenated wind-speed / classification cells (STREWS, merged D/G/H classes)	Verify published table before shingle or PV-shingle class; footnote 18 m height assumption is readable and typically exceeded on a high-rise
Table 1504.9	168 flattened parapet-height cells (6 mean-height rows × 2 aggregate sizes × 14 wind columns as marked in the extract HTML)	Do not reconstruct millimetre rows; footnotes c–d (50 mm gravel stop, Exposure D + 200 mm / ≥ 300 mm) may be used only after the published cell is known
Table 1505.1	9 concatenated construction-type class tokens including degree-symbol C°	Do not adopt CS.md/commentary Class C mapping; verify the published table against construction type
1507.1.1 Exception 2	1 thickness token Ø . 25 with no unit	Verify published cap-thickness unit before specifying power-driven metal caps from the exception
Table 1507.1.1(1)	1 wind-speed header (V > 62.5 m/s and V ≥ 62.5 m/s both printed)	Verify which column is the low-wind vs high-wind type list
Table 1507.1.1(2)	36 flattened underlayment-application numbers with OCR (not forever than , slope-band mix-ups)	Prefer 1507.1.1 exception prose; verify published application table if steep-slope is added
Table 1507.3.7	35 concatenated tile-fastening cells	Use 1507.3.6 prose fasteners; verify published table before wind-speed nailing
Table 1507.4.3(1)	5 copper mass cells (Ø . 0416 / Ø . 0312 kg/m²) inconsistent with 1505.2 4.9 kg/m² copper	Verify published copper masses; aluminium 0.61 / 0.48 mm and zinc 0.7 mm remain usable
Table 1507.4.3(2)	1 coating-row OCR (505% vs 5% aluminium-alloy)	Verify published corrosion-resistance row
Table 1507.8	2 spacing tokens (6.4 to 9.5 mm) that conflict with 1507.8.7 6 to 9 mm	Use the clause spacing; verify the published installation table
Table 1507.8.7	18 rowspan-concatenated weather-exposure cells	Verify published shingle exposure before cutting shingles
Table 1507.9.8	12 concatenated shake-exposure cells	Verify published shake exposure; footnote a (600 mm × 9.5 mm, 190 mm) is readable
1511.2.4 Exception 3	4 unreadable Type III/IV/V penthouse-wall limits (repeated OCR, page split)	Verify published Exception 3; do not use Type I Exception 1 as a substitute for III–V
1511.3–1511.3.2	Page-split: tank clauses missing after Exception 3 OCR; no numeric tank limits recovered from the extract	Verify published 1511.3–1511.3.2; do not promote commentary about tanks over stairs/elevators
1511.5.1	1 building-height token 1.5 m adjacent to 15 m in the same paragraph	Verify published trigger; keep 18 m, 18.6 m², 1.5-hour and 15 m as printed elsewhere in the clause